

Phase 4 — Validation & Pipeline Engineering

Part A: Testing Pyramid (70% Unit / 20% Integration / 10% E2E)

Test Distribution Summary

Layer	Target	Count	Modules Covered
Unit	70%	49	Auth, Menu & Cart, Order & Payment, Stock, Lifecycle
Integration	20%	14	Cross-module API flows, DB transactions
E2E	10%	7	Full user journeys via Playwright
Total	100%	70	All 5 modules

Unit Tests (49 tests — 70%)

Module 1: Auth & Identity (10 unit tests)

[illegible]

Module 2: Menu & Cart (10 unit tests)

```
# tests/unit/test_menu_cart_unit.py # ■■ FR14-16: Voucher Validation ■■■ class TestVoucherUnit: def test_voucher_discount_floors_at_zero(self): """Unit: final_total = max(0, total - discount).""" cart_total = 30.00 discount = 100.00 final = max(0, cart_total - discount) assert final == 0.00 assert final >= 0.00
```

Module 3: Order & Payment (12 unit tests)

Module 4: Stock & Resilience (9 unit tests)

Module 5: Lifecycle & Reports (8 unit tests)

[illegible]

Integration Tests (14 tests — 20%)

[illegible]

E2E Tests (7 tests — 10%)

E2E tests run via **Playwright** against the full stack (React frontend + Flask/FastAPI backend + PostgreSQL). See **Part B** for full Playwright scripts with Page Object Model.

#	E2E Scenario	FRs Covered
1	Student login → browse menu → place order	FR01, FR09, FR20

2	Payment flow: select method → confirm	FR26, FR27
3	Account lockout after 5 failed logins	FR03
4	Session expiry redirects to login	FR04
5	Admin creates user + suspends account	FR50, FR51
6	Password reset request flow	FR06
7	Order cancellation within window	FR37

Test Report Summary

Metric	Value
Total Test Count	70
Unit Tests	49 (70.0%)
Integration Tests	14 (20.0%)
E2E Tests	7 (10.0%)
Pyramid Ratio	70 / 20 / 10 
Modules Covered	5 / 5 (100%)
Edge Cases Covered	24 / 24 (100%)
Hidden Requirements	HR-01 to HR-07 (100%)
Framework (Unit)	pytest
Framework (Integration)	pytest + httpx/TestClient
Framework (E2E)	Playwright